

Experiment 5

Module Name: Robot Frameworks

Experiment Name: Test Setup and Teardown

Problem Statement: Set up a test suite with a common test setup that opens a browser and logs in before each test case. Define a test teardown that logs out or cleans up resources after each test case

Objective: To modularize repetitive state-management tasks (pre-conditions and post-conditions) using Setup and Teardown configurations.

Tools, Software, and Concepts Used:

- Test Setup and Test Teardown configuration hooks
- Selenium Library

Source Code / Implementation Steps:

1. Create a folder inside that folder create 'Experiment5.robot' and write this:

```
*** Settings ***
Library           SeleniumLibrary
Test Setup        Launch App And Login
Test Teardown     Logout And Close Browser

*** Variables ***
${URL}            https://www.saucedemo.com/
${BROWSER}        Chrome

*** Keywords ***
Launch App And Login
    Open Browser    ${URL}        ${BROWSER}
    Input Text      id=user-name    standard_user
    Input Text      id=password     secret_sauce
    Click Button    id=login-button

Logout And Close Browser
    Click Button    id=react-burger-menu-btn
    Wait Until Element Is Visible    id=logout_sidebar_link
    Click Link      id=logout_sidebar_link
    Close Browser

*** Test Cases ***
Verify Shopping Cart Access
    [Documentation]    Validates cart after automatic login.
    Click Link        class=shopping_cart_link
    Page Should Contain Element    class=cart_contents_container

Verify Product Sorting
    [Documentation]    Validates sorting after automatic login.
    Select From List By Value    class=product_sort_container    za
```

2. Now from the folder go to cmd and write this:

```
>>> python -m robot Experiment5.robot
```

Output with clearly labelled screenshots:

```
D:\My Desktop\Robot Project>python -m robot Experiment5.robot
=====
Experiment5
=====
Verify Shopping Cart Access :: Validates cart after automatic login. | PASS |
=====
Experiment5 | PASS |
1 test, 1 passed, 0 failed
=====
Output: D:\My Desktop\Robot Project\output.xml
Log:    D:\My Desktop\Robot Project\log.html
Report: D:\My Desktop\Robot Project\report.html
```

Screenshot 1: Terminal Output

Experiment5 Report

Generated: 20260921 00:35:00 UTC+05:30
1 minute 21 seconds ago

Summary

Status: All tests passed
Start Time: 20260921 00:34:53.493
End Time: 20260921 00:35:00.017
Elapsed Time: 00:00:06.524
Log File: [log.html](#)

Statistics

Total Statistics	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
All Tests	1	1	0	0	00:00:06	

Statistics by Tag	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
No Tags						

Statistics by Suite	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
Experiment5	1	1	0	0	00:00:07	

Details

All Tags Suites Search

Suite:
Test:
Include:
Exclude:

Search Clear Help

Screenshot 2: Output of report.html

Brief Result: The browser successfully launched, logged in, executed the unique test steps, logged out, and closed down identically for both test cases.

Observation: The setup and teardown steps do not clutter the `*** Test Cases ***` section, keeping the actual test logic highly readable.

Conclusion: Using Setups and Teardowns ensures isolated, independent test execution, preventing a failure in one test case from causing cascading failures in subsequent ones.